

Experiment - 2

TestDisk: Open-Source Data Recovery Tool

recover a missing partition and repair a corrupted one using test disk

Installation

Linux : `sudo apt-get install testdisk`

macOS : `brew install testdisk`

Windows: Download the executable from the official website.

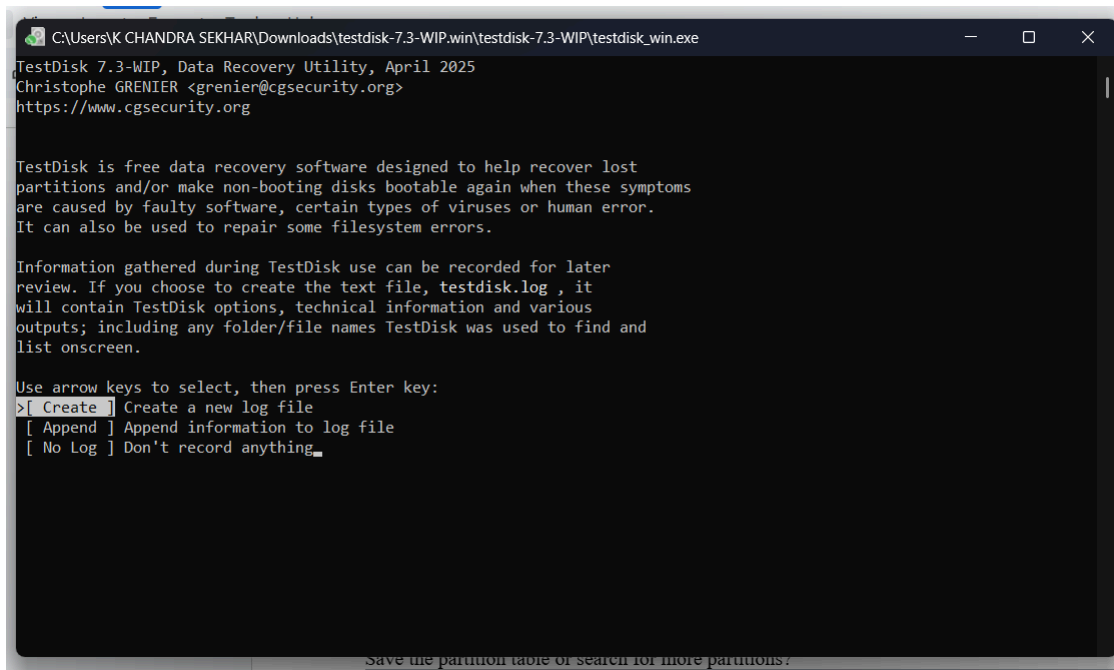
Link : https://www.cgsecurity.org/wiki/TestDisk_Download

NOTE:- if you using in windows please extract the .zip file first

Procedure

1. Create a Log File

- Launch TestDisk from your terminal or command prompt using `sudo testdisk` (or `testdisk_win.exe` on Windows).
- Select the [Create] option to generate a log file of the recovery session. This is helpful for future reference or troubleshooting.



2. Select the Drive to Analyze

- TestDisk will display a list of all detected storage devices.
- Use the Up/Down arrow keys to highlight the drive that contains your lost data.
- Select [Proceed] to move to the next step.

```
C:\Users\K CHANDRA SEKHAR\Downloads\testdisk-7.3-WIP.win\testdisk-7.3-WIP\testdisk_win.exe
TestDisk 7.3-WIP, Data Recovery Utility, April 2025
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
Disk \\.\PhysicalDrive0 - 512 GB / 476 GiB - Phison ES0512GHLCA1-E9C
>Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - SDXC

>[Proceed] [Quit]

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
Save the partition table or search for more partitions?
```

3. Choose the Partition Table Type

- TestDisk will automatically suggest the most likely partition table type (e.g., Intel/PC, EFI GPT).
- The default value is usually correct. Confirm the selection by pressing Enter.

```
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - SDXC

Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

4. Analyze Current Partition Structure

- From the main menu, choose [Analyse] and press Enter.
- This will display the current partition table and check for errors or missing partitions.

```
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - SDXC

Please select the partition table type, press Enter when done.
>[Intel ] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax ] Humax partition table
[Mac ] Apple partition map (legacy)
[None ] Non partitioned media
[Sun ] Sun Solaris partition
[XBox ] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.

Save the partition table or search for more partitions?
```

5. Perform a Quick Search

- After the analysis, you will be prompted to perform a [Quick Search].
- Select it and press Enter to scan the drive for lost partitions.

```
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63
Current partition structure:
  Partition      Start      End      Size in sectors
  1 P HPFS - NTFS    2 10 9    7791 180 5    125140992
No partition is bootable

*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
>[Quick Search] [ Backup ]
Try to locate partition.
Save the partition table or search for more partitions?
```

-
- Once a partition is found, you can press p to list its files. Deleted files and folders often appear in red.
- Press q to return to the search results.

6. Perform a Deeper Search

- If the Quick Search fails to find your lost partitions, select [Deeper Search]
- This process can take a long time, as it scans the entire drive, block by block, to find remnants of partition structures.

```
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63
Partition      Start      End      Size in sectors
>* HPFS - NTFS  2 10 9 7791 180 5 125140992

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
*=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
exFAT, blocksize=131072, 64 GB / 59 GiB
```

- Again, use p to preview files and confirm if a found partition is the one you are looking for.

7. Modify Partition Status

- After finding the correct partitions, use the Left/Right arrow keys to set their status.

```
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63

Partition              Start          End      Size in sectors
1 P HPFS - NTFS         2  10  9  7791 180  5  125140992

[ Quit ] [ Return ] >[Deeper Search] [ Write ]
                          Try to find more partitions
```

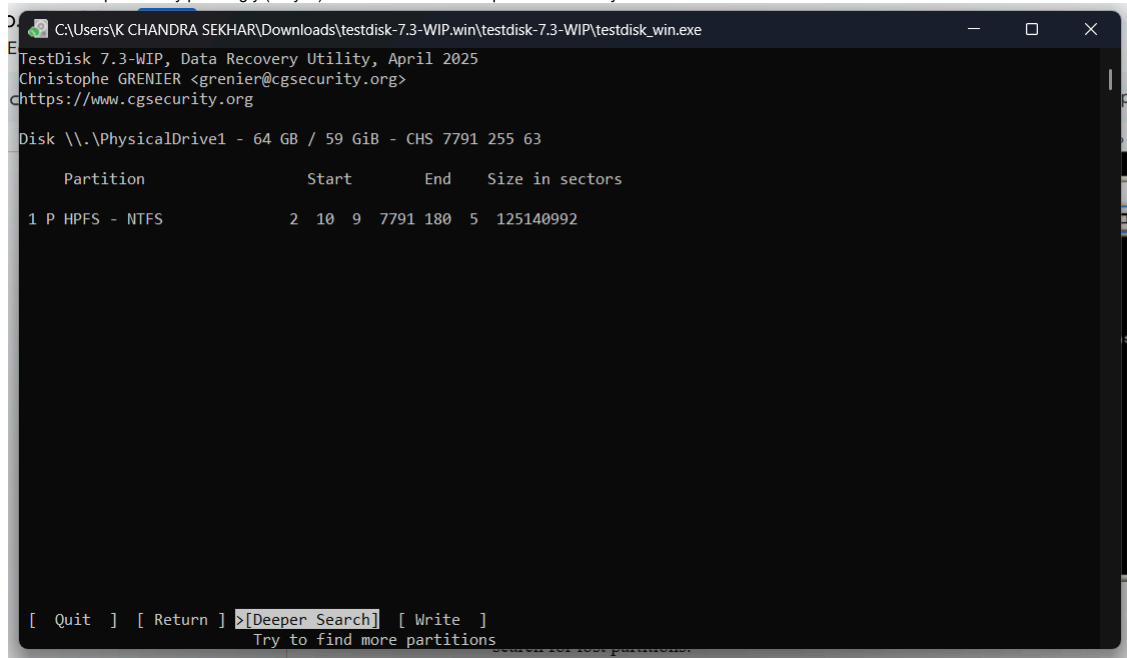
• Use Left/Right arrow keys to change status:

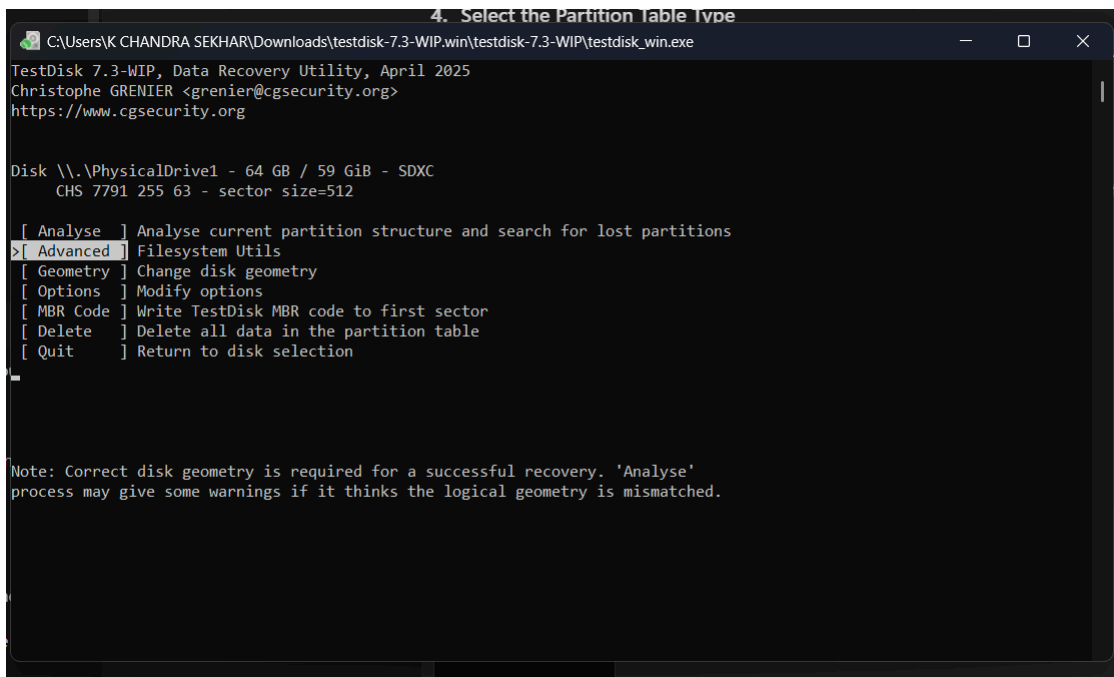
- P: Primary
- *: Bootable
- L: Logical
- D: Deleted
- ensure that the partitions you want to recover are marked as Primary or Logical (and not deleted).

8. Write the Partition Table

- Once you are confident the partition structure is correct, select [Write] from the menu.

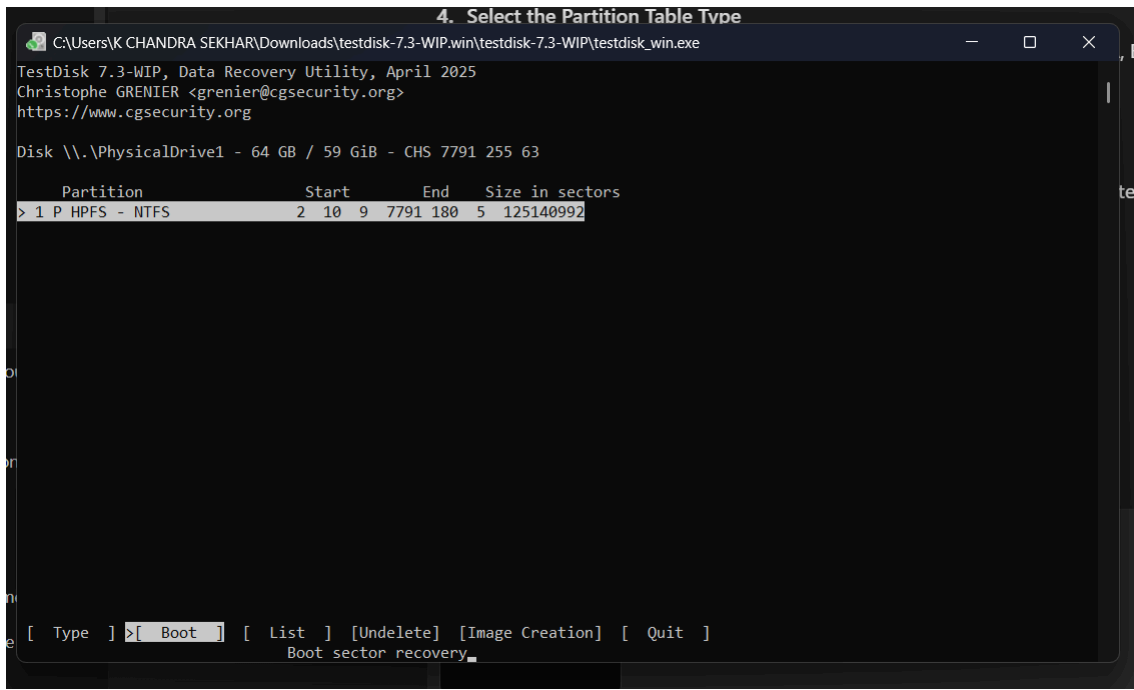
- Confirm the operation by pressing y (for yes). This will write the new partition table to your disk.





9. Recover Files

- If you just need to recover a few files without fixing the partition table, you can do so from the file list (after pressing p).
- Navigate to the folder containing your desired files.
- Use the colon : key to select the files you want to recover.
- Press the uppercase C key to copy the selected file(s).
- Navigate to a safe destination on a different storage device and press uppercase C again to paste.



10. Exit and Restart

- Once your task is complete, select [Quit] to exit the program.
- If you wrote a new partition table to the drive, it is recommended to restart your computer to allow the operating system to recognize the changes.

```
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Disk \\.\PhysicalDrive1 - 64 GB / 59 GiB - CHS 7791 255 63
Partition      Start      End      Size in sectors
1 P HPFS - NTFS      2  10  9  7791 180  5  125140992

Boot sector
exFAT OK

Backup boot record
exFAT OK

Sectors are not identical.

>[ Quit ]  [Org. BS ]  [Backup BS]  [ Dump ]
          Return to Advanced menu
```

References:

https://www.cgsecurity.org/wiki/TestDisk_Download

https://www.cgsecurity.org/testdisk_doc/